Question	Answers	Extra information	Mark	AO / Spec Ref.
04.7	each person's sense of taste is different		1	AO3 4.2.2.1

Total Question 4	**14**

Question 5

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.1	ionising radiation viruses		1 1	AO1 4.2.2.5 4.2.2.6

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.2	mitosis		1	AO1 4.2.2.7 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.3	grow replicate	must be in this order	1 1	AO1 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.4	40%		1	AO2 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.5	chromosomes are pulled to each end of the cell		1	AO2 4.1.2.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.6	(cell) membrane		1	AO2 4.1.1.2

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.7		allow conversion to µm at any stage in the calculation		AO2 4.1.1.5
	$\frac{50}{800}$		1	
	0.0625 (mm)		1	
	62.5 (µm)	allow correct conversion from student's incorrect calculation	1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.8		must be in this order		AO1 4.3.1.9
	cells		1	
	people		1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
05.9	placebo	allow placebo effect ignore examples and descriptions of placebos	1	AO1 4.3.1.9

Total Question 5	**14**

Question 6

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.1	any **two** from: • size of piece of potato • the (type of) potato • volume / 100 cm^3 of salt solution • time (pieces of potato are kept) in the solution / beaker • the potato was uncooked	ignore size of potato allow amount of salt solution allow 20 minutes (the pieces of potato are kept) in the solution / beaker ignore time unqualified	2	AO1 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.2	blot **or** dry (the surface)	allow descriptions of blotting allow descriptions of drying (the surface)	1	AO3 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.3	balance **or** weighing scale		1	AO1 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.4	0.1 g		1	AO2 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.5	D		1	AO3 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.6	$\frac{1.1}{6.0} \times 100$	ignore minus sign throughout	1	AO2 4.1.3.2 RPA3
	18.333…		1	
	18.3(%)	allow correct conversion to 1 decimal place from student's incorrect calculation using figures from potato piece **D**	1	

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.7	line graph		1	AO2 4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.8		must be in this order		
	water		1	AO2
	osmosis	allow diffusion	1	AO1
	permeable (membrane)		1	AO2
				4.1.3.2 RPA3

Question	Answers	Extra information	Mark	AO / Spec Ref.
06.9	answer in the range 0.15 to 0.25 (mol/dm^3)		1	AO3 4.1.3.2 RPA3

Total Question 6	**14**